Reliable Rentals

Database Implementation Report (Part 3)

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1. Introduction

This report presents the Part 3 implementation of the Reliable Rentals database using embedded SQL with Python and SQLite.

The goal of this stage is to convert the logical database design produced in Part 2 into a working physical database that can be created, populated, queried, and verified through executable code.

The implementation preserves the business structure established in Parts 1 and 2.

The five main business relations remain Outlet, Staff, Vehicle, Client, and HireAgreement.

The final physical implementation also includes one auxiliary support table named OutletStaffCoverage.

This support table is not treated as a new business entity from the conceptual model.

Instead, it is included as a physical implementation mechanism to enforce the rule that every outlet must always retain at least one staff member.

2. Implementation Method

The database is implemented in the Python script group assigmment-part3.py using Python's sqlite3 library.

The SQL statements are embedded directly inside the Python program as multiline SQL strings.

This satisfies the assignment requirement to develop SQL code using Python and SQLite as the host environment.

When the script is executed, it deletes any previous copy of the SQLite database and rebuilds the database from scratch.

This ensures that each run starts from a clean state and produces repeatable results.

The script then creates the schema, inserts sample data, displays the contents of the relations, and executes the five required user transaction queries from Part 2.C.

3. Database Schema and Constraints

The physical database contains the major relations identified during conceptual and logical design.

Outlet stores the details of each rental branch, including outlet number, address, phone number, and optional fax number.

Staff stores employee data such as name, address, contact number, date of birth, sex, date joined, job title, salary, and assigned outlet.

Vehicle stores the registration number, model, make, engine size, seating capacity, current mileage, daily hire rate, and current outlet.

Client stores each customer's identifier, name, address, phone number, date of birth, and driving license number.

HireAgreement stores the agreement identifier, start and end dates, mileage values, historical snapshot values, and the foreign keys linking the agreement to the client and vehicle.

The additional support relation OutletStaffCoverage is used only to enforce the minimum staffing business rule at the database level.

3.1 Keys and Referential Integrity

Each relation has a clearly defined primary key.

Outlet uses OutletNumber as its primary key.

Staff uses StaffNumber as its primary key.

Vehicle uses RegistrationNumber as its primary key.

Client uses ClientNumber as its primary key.

HireAgreement uses HireNumber as its primary key.

Client.DrivingLicenseNumber is enforced as an alternate key through a UNIQUE constraint.

Foreign key relationships are implemented directly in the physical schema.

Each staff member references exactly one outlet.

Each vehicle references exactly one outlet.

Each hire agreement references exactly one client and exactly one vehicle.

Referential integrity is enforced through SQLite foreign key constraints together with PRAGMA foreign_keys = ON.

3.2 Domain and Business Constraints

The implementation includes domain checks to ensure that values entered into the database remain valid.

Phone and fax values are restricted to 11-digit numeric strings.

Required text fields are protected against null or blank values.

Salary, engine size, and daily hire rate must be positive values.

Capacity must be at least 1, and mileage values cannot be negative.

Vehicle registration numbers are explicitly enforced as NOT NULL as well as PRIMARY KEY.

EndDate must be equal to or later than StartDate when present.

MileageAfter must be equal to or greater than MileageBefore when present.

Several business rules are implemented through triggers.

These triggers validate staff dates and client birth dates, prevent overlapping hire periods for the same vehicle, and enforce the requirement that every outlet must retain at least one staff member.

Screenshot 1. Schema Objects Created

4. Sample Data Population

The script inserts sample data into the main business relations in the database.

Five outlet rows are inserted into Outlet.

Five staff rows are inserted into Staff.

Five vehicle rows are inserted into Vehicle.

Five client rows are inserted into Client.

Five hire agreement rows are inserted into HireAgreement.

This satisfies the assignment requirement to create at least five tuples for each main relation in the database.

Screenshot 2. Outlet Table Contents

Outlet Table Contents

```
SELECT * FROM Outlet ORDER BY OutletNumber;
+----------------+-----------------------------+---------------+-------------+
|   OutletNumber | Address                     |   PhoneNumber |   FaxNumber |
+================+=============================+===============+=============+
|              1 | 101 King Street, Glasgow    |   01415550101 | 01415550199 |
+----------------+-----------------------------+---------------+-------------+
|              2 | 22 Princes Street, Edinburgh |   01315550202 | 01315550299 |
+----------------+-----------------------------+---------------+-------------+
|              3 | 8 Union Square, Aberdeen    |   01224550303 |        NULL |
+----------------+-----------------------------+---------------+-------------+
|              4 | 55 Tay Road, Dundee         |   01382550404 |        NULL |
+----------------+-----------------------------+---------------+-------------+
|              5 | 14 High Street, Stirling    |   01786550505 | 01786550599 |
+----------------+-----------------------------+---------------+-------------+
```

Figure 1: Outlet Table Contents

Screenshot 3. Staff Table Contents

Staff Table Contents

```
SELECT * FROM Staff ORDER BY StaffNumber;
+---------------+-------------+-----------+-------------------------+-----------------+-----------+-------+-------------+-------------------+---------+----------------+
|   StaffNumber | FirstName   | LastName  | HomeAddress             |  HomePhoneNumber | DOB       | Sex   | DateJoined  | JobTitle          |  Salary |  OutletNumber |
+===============+=============+===========+=========================+=================+===========+=======+=============+===================+=========+================+
|             1 | Maria       | Lopez     | 12 Birch Road, Glasgow  |      07700900101 | 1988-05-14 | F     | 2018-03-01  | Manager           |   42000 |              1 |
+---------------+-------------+-----------+-------------------------+-----------------+-----------+-------+-------------+-------------------+---------+----------------+
|             2 | David       | Chen      | 44 Elm Street, Edinburgh |      07700900202 | 1990-09-21 | M     | 2019-06-15  | Assistant Manager |   36500 |              2 |
+---------------+-------------+-----------+-------------------------+-----------------+-----------+-------+-------------+-------------------+---------+----------------+
|             3 | Priya       | Patel     | 9 Cedar Avenue, Aberdeen |      07700900303 | 1992-02-08 | F     | 2020-01-20  | Rental Agent      |   29500 |              3 |
+---------------+-------------+-----------+-------------------------+-----------------+-----------+-------+-------------+-------------------+---------+----------------+
|             4 | James       | Walker    | 71 Pine Close, Dundee   |      07700900404 | 1985-11-30 | M     | 2017-08-11  | Mechanic          |   33200 |              4 |
+---------------+-------------+-----------+-------------------------+-----------------+-----------+-------+-------------+-------------------+---------+----------------+
|             5 | Sarah       | Ahmed     | 3 Willow Lane, Stirling |      07700900505 | 1994-07-12 | F     | 2021-04-05  | Rental Agent      |   28750 |              5 |
+---------------+-------------+-----------+-------------------------+-----------------+-----------+-------+-------------+-------------------+---------+----------------+
```

Figure 2: Staff Table Contents

Vehicle Table Contents

```
SELECT * FROM Vehicle ORDER BY RegistrationNumber;
+----------------------+---------+---------+--------------+------------+------------------+-----------------+----------------+
| RegistrationNumber   | Model   | Make    |   EngineSize |   Capacity |   CurrentMileage |   DailyHireRate |   OutletNumber |
+======================+=========+=========+==============+============+==================+=================+================+
| ABD-3003             | CR-V    | Honda   |          2   |          5 |            31000 |              72 |              3 |
+----------------------+---------+---------+--------------+------------+------------------+-----------------+----------------+
| DUN-4004             | Micra   | Nissan  |          1.2 |          5 |            12200 |              39 |              4 |
+----------------------+---------+---------+--------------+------------+------------------+-----------------+----------------+
| EDI-2002             | Focus   | Ford    |          1.5 |          5 |            18300 |              46 |              2 |
+----------------------+---------+---------+--------------+------------+------------------+-----------------+----------------+
| GLA-1001             | Corolla | Toyota  |          1.8 |          5 |            24500 |              48 |              1 |
+----------------------+---------+---------+--------------+------------+------------------+-----------------+----------------+
| GLA-1006             | Tucson  | Hyundai |          2   |          5 |             8700 |              74 |              1 |
+----------------------+---------+---------+--------------+------------+------------------+-----------------+----------------+
```

Figure 3: Vehicle Table Contents

Screenshot 4. Vehicle Table Contents

Screenshot 5. Client Table Contents

Client Table Contents

```
SELECT * FROM Client ORDER BY ClientNumber;
+----------------+-------------+------------+---------------------------+---------------+---------------+------------------------+
|   ClientNumber | FirstName   | LastName   | HomeAddress               |   PhoneNumber | DateOfBirth   | DrivingLicenseNumber   |
+================+=============+============+===========================+===============+===============+========================+
|              1 | Emma        | Johnson    | 9 Rose Street, Glasgow    |   07800111111 | 1995-01-17    | DLA123456A             |
+----------------+-------------+------------+---------------------------+---------------+---------------+------------------------+
|              2 | Noah        | Smith      | 27 Castle Road, Edinburgh |   07800222222 | 1989-06-02    | DLB234567B             |
+----------------+-------------+------------+---------------------------+---------------+---------------+------------------------+
|              3 | Olivia      | Brown      | 18 Harbour View, Aberdeen |   07800333333 | 1993-09-25    | DLC345678C             |
+----------------+-------------+------------+---------------------------+---------------+---------------+------------------------+
|              4 | Liam        | Wilson     | 63 Riverside Drive, Dundee |   07800444444 | 1987-12-11    | DLD456789D             |
+----------------+-------------+------------+---------------------------+---------------+---------------+------------------------+
|              5 | Ava         | Taylor     | 5 Meadow Lane, Stirling   |   07800555555 | 1998-04-07    | DLE567890E             |
+----------------+-------------+------------+---------------------------+---------------+---------------+------------------------+
```

Figure 4: Client Table Contents

Screenshot 6. HireAgreement Table Contents

HireAgreement Table Contents

```
SELECT * FROM HireAgreement ORDER BY HireNumber;
Split layout used to improve readability in the report.
```

Part A. Core agreement fields

HireNumber	StartDate	EndDate	MileageBefore	MileageAfter	ClientNumber	RegistrationNumber
1001	2026-01-10	2026-01-14	24500	24980	1	GLA-1001
1002	2026-02-05	2026-02-08	24980	25310	2	GLA-1001
1003	2026-03-01	2026-03-04	18300	18610	1	EDI-2002
1004	2026-03-15	2026-03-20	8700	9100	3	GLA-1006
1005	2026-04-01	NULL	12200	NULL	4	DUN-4004

Part B. Snapshot summary fields

HireNumber	ClientFirstNameSnapshot	ClientLastNameSnapshot	ClientPhoneSnapshot	VehicleModelSnapshot	VehicleMakeSnapshot
1001	Emma	Johnson	07800111111	Corolla	Toyota
1002	Noah	Smith	07800222222	Corolla	Toyota
1003	Emma	Johnson	07800111111	Focus	Ford
1004	Olivia	Brown	07800333333	Tucson	Hyundai
1005	Liam	Wilson	07800444444	Micra	Nissan

Part C. Client address snapshots

HireNumber	ClientAddressSnapshot
1001	9 Rose Street, Glasgow
1002	27 Castle Road, Edinburgh
1003	9 Rose Street, Glasgow
1004	18 Harbour View, Aberdeen
1005	63 Riverside Drive, Dundee

Figure 5: HireAgreement Table Contents

5. SQL Queries for the Required User Transactions

The Python implementation includes the five SQL queries required in Part 2.C.

Each query is executed using embedded SQL with parameterised placeholders.

This improves clarity and avoids unsafe string concatenation.

5.1 Transaction 1

Transaction 1 returns the vehicles currently stocked at a selected outlet.

The query joins Outlet and Vehicle through OutletNumber and returns the outlet details together with each vehicle's registration number, make, model, mileage, and daily rate.

Screenshot 7. Transaction 1 Output

Transaction 1: Vehicles stocked at outlet 1

```
SELECT o.OutletNumber, o.Address, v.RegistrationNumber, v.Make, v.Model,
               v.CurrentMileage, v.DailyHireRate
        FROM Outlet AS o
        JOIN Vehicle AS v ON v.OutletNumber = o.OutletNumber
        WHERE o.OutletNumber = ?
        ORDER BY v.RegistrationNumber;
Parameters: (1,)
+----------------+-------------------------+----------------------+---------+---------+------------------+-----------------+
|   OutletNumber | Address                 | RegistrationNumber   | Make    | Model   |   CurrentMileage |   DailyHireRate |
+================+=========================+======================+=========+=========+==================+=================+
|              1 | 101 King Street, Glasgow | GLA-1001             | Toyota  | Corolla |            24500 |              48 |
+----------------+-------------------------+----------------------+---------+---------+------------------+-----------------+
|              1 | 101 King Street, Glasgow | GLA-1006             | Hyundai | Tucson  |             8700 |              74 |
+----------------+-------------------------+----------------------+---------+---------+------------------+-----------------+
```

Figure 6: Transaction 1 Output

5.2 Transaction 2

Transaction 2 returns the staff members working at a selected outlet.

The query joins Outlet and Staff and returns staff number, name, job title, date joined, and salary.

Screenshot 8. Transaction 2 Output

Transaction 2: Staff working at outlet 1

```
SELECT o.OutletNumber, s.StaffNumber, s.FirstName, s.LastName,
               s.JobTitle, s.DateJoined, s.Salary
        FROM Outlet AS o
        JOIN Staff AS s ON s.OutletNumber = o.OutletNumber
        WHERE o.OutletNumber = ?
        ORDER BY s.StaffNumber;
Parameters: (1,)
+----------------+---------------+-------------+------------+------------+--------------+----------+
|   OutletNumber |   StaffNumber | FirstName   | LastName   | JobTitle   | DateJoined   |   Salary |
+================+===============+=============+============+============+==============+==========+
|              1 |             1 | Maria       | Lopez      | Manager    | 2018-03-01   |    42000 |
+----------------+---------------+-------------+------------+------------+--------------+----------+
```

Figure 7: Transaction 2 Output

5.3 Transaction 3

Transaction 3 returns the vehicles hired by a selected client and the dates of those hire agreements. The query joins Client, HireAgreement, and Vehicle to show the hire history of the selected client.

Screenshot 9. Transaction 3 Output

Transaction 3: Vehicles hired by client 1

```
SELECT c.ClientNumber, c.FirstName, c.LastName, h.HireNumber,
               h.StartDate, h.EndDate, v.RegistrationNumber, v.Make, v.Model
        FROM Client AS c
        JOIN HireAgreement AS h ON h.ClientNumber = c.ClientNumber
        JOIN Vehicle AS v ON v.RegistrationNumber = h.RegistrationNumber
        WHERE c.ClientNumber = ?
        ORDER BY h.StartDate;
Parameters: (1,)
+----------------+-------------+------------+--------------+-------------+------------+----------------------+--------+---------+
|   ClientNumber | FirstName   | LastName   |   HireNumber | StartDate   | EndDate    | RegistrationNumber   | Make   | Model   |
+================+=============+============+==============+=============+============+======================+========+=========+
|              1 | Emma        | Johnson    |         1001 | 2026-01-10  | 2026-01-14 | GLA-1001             | Toyota | Corolla |
+----------------+-------------+------------+--------------+-------------+------------+----------------------+--------+---------+
|              1 | Emma        | Johnson    |         1003 | 2026-03-01  | 2026-03-04 | EDI-2002             | Ford   | Focus   |
+----------------+-------------+------------+--------------+-------------+------------+----------------------+--------+---------+
```

Figure 8: Transaction 3 Output

5.4 Transaction 4

Transaction 4 returns the clients who hired a specified vehicle together with the corresponding hire periods. The query joins Vehicle, HireAgreement, and Client to retrieve the full history of hires for that vehicle.

Screenshot 10. Transaction 4 Output

Transaction 4: Clients who hired vehicle GLA-1001

```
SELECT v.RegistrationNumber, h.HireNumber, h.StartDate, h.EndDate,
               c.ClientNumber, c.FirstName, c.LastName, c.DrivingLicenseNumber
        FROM Vehicle AS v
        JOIN HireAgreement AS h ON h.RegistrationNumber = v.RegistrationNumber
        JOIN Client AS c ON c.ClientNumber = h.ClientNumber
        WHERE v.RegistrationNumber = ?
        ORDER BY h.StartDate;
Parameters: ('GLA-1001',)
+----------------------+--------------+-------------+------------+----------------+-------------+------------+------------------------+
| RegistrationNumber   |   HireNumber | StartDate   | EndDate    |   ClientNumber | FirstName   | LastName   | DrivingLicenseNumber   |
+======================+==============+=============+============+================+=============+============+========================+
| GLA-1001             |         1001 | 2026-01-10  | 2026-01-14 |              1 | Emma        | Johnson    | DLA123456A             |
+----------------------+--------------+-------------+------------+----------------+-------------+------------+------------------------+
| GLA-1001             |         1002 | 2026-02-05  | 2026-02-08 |              2 | Noah        | Smith      | DLB234567B             |
+----------------------+--------------+-------------+------------+----------------+-------------+------------+------------------------+
```

Figure 9: Transaction 4 Output

5.5 Transaction 5

Transaction 5 returns the details of a specific hire agreement together with the related client, vehicle, and current outlet.

The query joins HireAgreement, Client, Vehicle, and Outlet so that the agreement can be viewed in its full operational context.

Screenshot 11. Transaction 5 Output

Transaction 5: Details for hire agreement 1003

```
Parameters: (1003,)
Split layout used to improve readability in the report.
Part A. Hire agreement details
+--------------+-------------+------------+-----------------+----------------+
|   HireNumber | StartDate   | EndDate    |   MileageBefore |   MileageAfter |
+==============+=============+============+=================+================+
|         1003 | 2026-03-01  | 2026-03-04 |           18300 |          18610 |
+--------------+-------------+------------+-----------------+----------------+

Part B. Client and vehicle details
+----------------+-------------+------------+---------------+----------------------+--------+---------+
|   ClientNumber | FirstName   | LastName   |   PhoneNumber | RegistrationNumber   | Make   | Model   |
+================+=============+============+===============+======================+========+=========+
|              1 | Emma        | Johnson    |   07800111111 | EDI-2002             | Ford   | Focus   |
+----------------+-------------+------------+---------------+----------------------+--------+---------+

Part C. Outlet details
+----------------+------------------------------+---------------+
|   OutletNumber | Address                      |   OutletPhone |
+================+==============================+===============+
|              2 | 22 Princes Street, Edinburgh |   01315550202 |
+----------------+------------------------------+---------------+
```

Figure 10: Transaction 5 Output

6. Execution and Verification

When the script is executed, it creates a new SQLite database file named reliable_rentals_part3.db in the same folder as the script.

The script prints the schema objects, table contents, and transaction query results to the terminal.

A clean copy of the execution output was also preserved in a text file for documentation and screenshot purposes.

The successful output demonstrates that the schema can be built correctly, the sample data can be inserted successfully, and all five required user transaction queries can be executed as intended.

7. GitHub Repository Submission

The final GitHub repository should include all materials from Parts 1, 2, and 3.

The repository should contain the reports, diagrams, implementation code, screenshot evidence, and a README file describing the repository structure and execution method.

For Part 3, the repository should include the final Python implementation script, the final report, and the screenshot files used as evidence in the report.

Repository URL: https://github.com/Royy26/dsc623-reliable-rentals-project

8. Conclusion

This part of the project successfully implements the Reliable Rentals database in SQLite using embedded SQL with Python.

The implementation reflects the logical design developed in Part 2 and preserves the conceptual structure established in Part 1.

The final database includes the required relations, valid sample data, and the five user transaction queries required by the assignment.

The implementation also enforces key integrity rules through primary keys, foreign keys, check constraints, and triggers.

Overall, the Part 3 implementation demonstrates that the database design can be translated from the conceptual and logical levels into a functioning and verifiable physical database system.